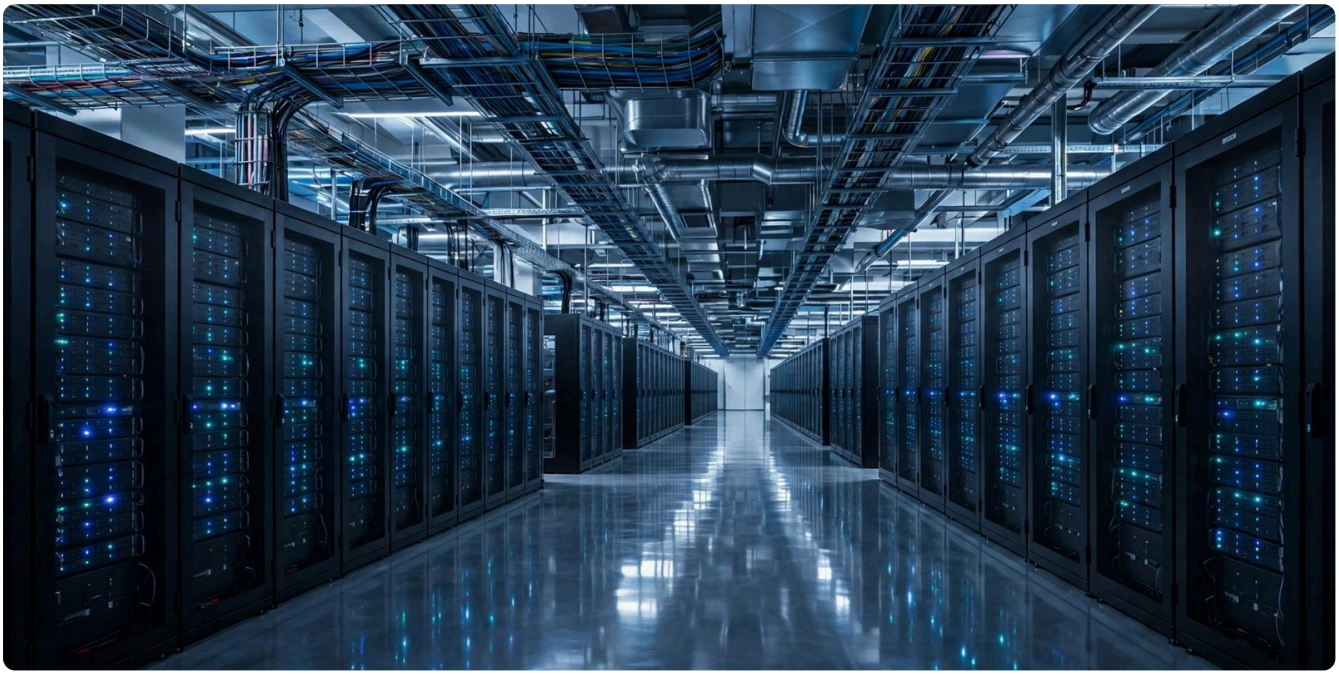
REFERENCE PACK - 11 PAGES - OCTOBER 2026

AI Data Center Documents

Power, cooling, efficiency and operations - explained with real published figures, dated and sourced. No simulated or demo data.



415 TWh

Global data centre electricity use in 2024 - IEA, Energy and AI (April 2025)

945 TWh

IEA base-case projection for 2030 - more than double 2024

1.56

Average PUE in 2024 - Uptime Institute Global Survey 2024

What this pack is: A printable set of documents you can read, share and use as templates - an executive summary, the global and U.S. pictures, how an AI facility works, efficiency measures, facility profiles, and operations checklists.
What it is not: A live monitor. No facility telemetry feed is connected, so nothing in this pack claims to be a live reading.

Cover image is an illustration, not a photograph of a specific facility. All statistics are from the published sources listed on page 10.

Contents and how to use this pack

Each document has one job. Use the summaries to decide, the checklists to act, and the sources page to verify.

Page	Document	Use it to
3	**Executive summary**	Get the whole story in two minutes
4	**The global picture**	Understand world demand and where it grows
5	**The United States picture**	See U.S. use in 2023-2024 and the 2028-2030 outlook
6	**Inside an AI facility**	Follow power and heat from grid to GPU rack
7	**Efficiency: PUE and beyond**	Judge efficiency honestly, not by one number
8	**Facility profiles**	Compare Edge, Regional and Campus designs
9	**Operations documents**	Run daily rounds, changes and incidents
10	**Sources, glossary, live-feed readiness**	Verify every figure and prepare a real feed
11	**Company finance and employment links**	Fund the company and hire the team - official links

Document control

- **Owner:** The person who keeps the document current. Name one owner per document.
- **Review:** Check figures when a source publishes a new edition - IEA and LBNL update on their own cycles, not monthly.
- **Version:** Date every copy. A figure without a date and a source is not usable.
- **Rule for this pack:** Real published data only. If a number cannot be traced to page 10, it does not belong in the pack.

> **Reading tip:** TWh means terawatt-hours, a measure of total electricity used over time. MW and GW measure power at a moment in time - the size of the load, not the total consumed.

Demand is set to more than double by 2030

AI has turned data centers from a steady load into the fastest-growing large load on many grids. Power, cooling and grid access - not servers alone - now decide how much AI capacity can be built.

The five facts that matter

- **Global scale:** Data centers used about 415 TWh in 2024, about 1.5% of world electricity. The International Energy Agency (IEA) base case reaches about 945 TWh by 2030.
- **Growth rate:** The IEA puts data center demand growth at about 15% per year to 2030 - more than four times faster than total electricity demand growth.
- **AI is the driver:** Accelerated servers - the GPU-heavy systems that train and run AI models - are growing at about 30% per year in the IEA analysis and account for about half of the net increase.
- **U.S. weight:** The United States accounted for about 45% of global data center electricity use in 2024, followed by China at about 25% and Europe at about 15% (IEA).
- **Efficiency has stalled at the average:** Average Power Usage Effectiveness (PUE) was 1.56 in 2024 and has barely moved since about 2019, even though the best new facilities do far better (Uptime Institute).

What this means in practice

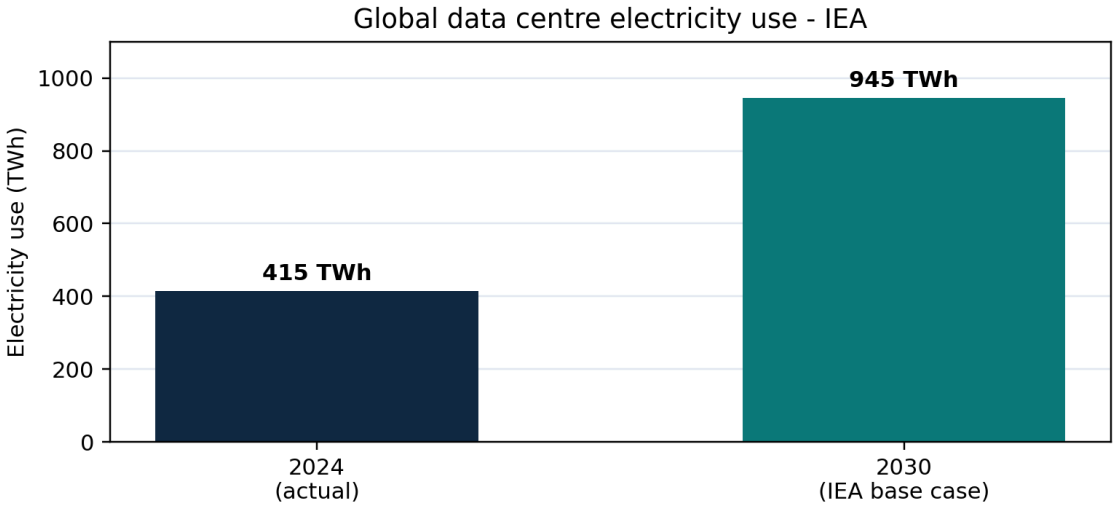
A new AI hall is planned backwards from power. First comes the grid connection and its queue time, then substation, switchgear and transformer capacity, then uninterruptible power supply (UPS) and backup generation, and only then the racks. Cooling is planned with the same care, because every watt of computing becomes heat that must be removed continuously.

For anyone reading a proposal, three questions cut through most claims: How many megawatts are secured, and from when? What PUE was measured, over what period, and at what load? And where does the heat and water go in the hottest week of the year?

> **Bottom line:** The opportunity is real and the constraint is physical. Treat electricity, cooling and lead times as the project - the computers are the part that arrives fastest.

The global picture: 415 TWh in 2024, 945 TWh by 2030

The IEA report *Energy and AI* (April 2025) is the clearest published global baseline. It counts all data centers - cloud, enterprise, colocation and AI-focused sites together - not AI alone.



~1.5%
Share of world electricity used by data centers in 2024

~15% / yr
Projected growth in data center demand to 2030 (IEA base case)

~30% / yr
Growth in accelerated (AI) server electricity use (IEA)

Where the electricity is used

Region	Share of global data center use, 2024	What to watch
United States	About 45%	Largest AI build-out; grid queues and local concentration
China	About 25%	Large domestic AI programs and rapid capacity additions
Europe	About 15%	Tighter efficiency and reporting rules; power constraints in hub cities
Rest of world	About 15%	Fast growth from a smaller base in Asia-Pacific and the Middle East

Concentration matters more than the global average suggests. Globally the share is still small, but in a single metro area or grid region a cluster of large facilities can become one of the biggest loads on the system, which is why local planning, not the world total, is where projects succeed or stall.

Source: IEA, Energy and AI, April 2025. The 2030 figure is a projection in the IEA base case, not a measured result.

The United States picture: from 4.4% toward a tenth of all power

The U.S. Department of Energy (DOE) work by Lawrence Berkeley National Laboratory (LBNL) is the standard U.S. reference. Its two recent editions use slightly different data, so this page keeps them separate and dated.

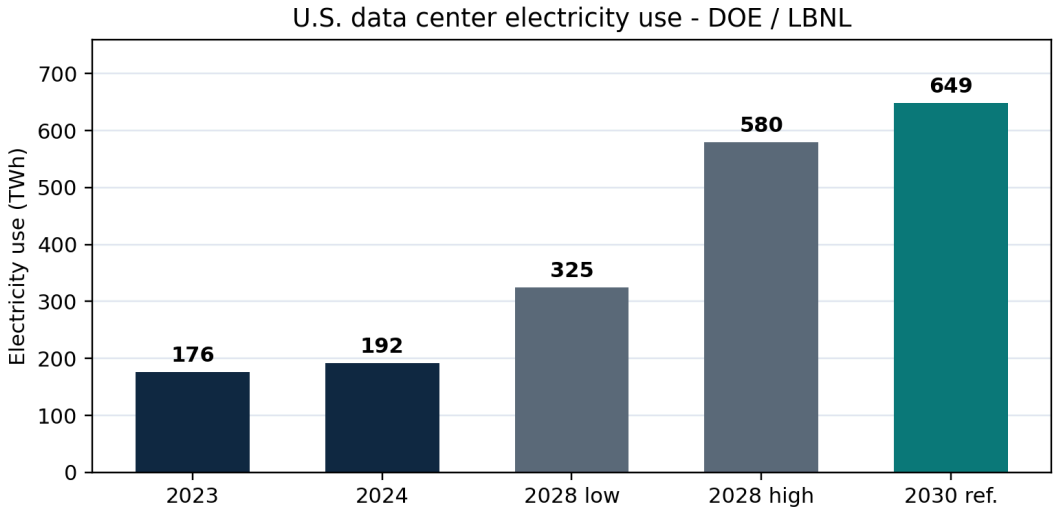


Figure	Value	Source edition
U.S. use, 2023	176 TWh - 4.4% of U.S. electricity	LBNL 2024 Report
U.S. use, 2024	192 TWh - 4.7% of U.S. electricity	LBNL 2025 Update
Projection, 2028	325-580 TWh - 6.7% to 12%	LBNL 2024 Report
Reference case, 2030	649 TWh - 11.8% of forecast U.S. use	LBNL 2025 Update
Scenario range, 2030	521-843 TWh - 9.5% to 15.3%	LBNL 2025 Update

Why the range is so wide

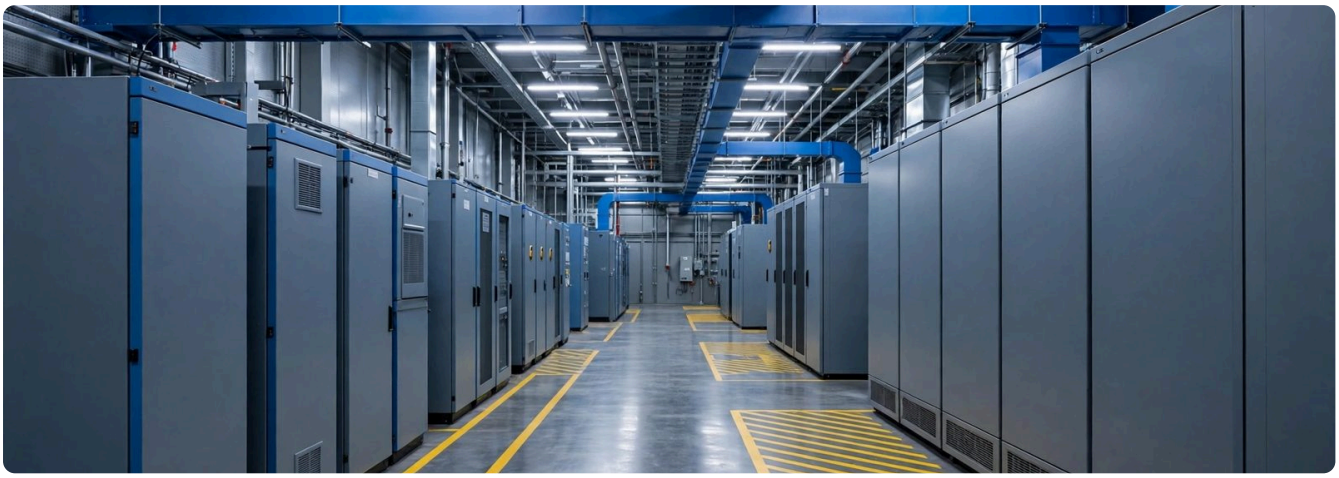
The range is honest uncertainty, not indecision. It depends on how many graphics processing units (GPUs) ship, how long AI servers stay in service, how inference - running models for users - scales, and how much announced capacity actually gets built and connected. Read any single-point forecast with that range in mind.

LBNL also notes what drives the increase: AI accelerators, not storage or ordinary web hosting. The practical bottleneck is often not generation alone, but transmission, substations, transformers and permitting, which move on a multi-year clock while chip orders move in months.

> **How to quote these figures:** Always give the year and the edition - for example, "192 TWh in 2024 (LBNL 2025 Update)". Mixing the 2024 and 2025 editions without labels produces numbers that appear to contradict each other.

Inside an AI facility: one chain, from grid to GPU

An AI data center is a power and heat system that happens to compute. If any link in the chain is undersized, the whole hall is undersized.



The power chain

- **Grid and substation:** One or two utility feeds, transformed down to facility voltage. Redundant feeds are the first resilience decision.
- **Switchgear and UPS:** Switchgear routes power; the UPS bridges outages for seconds to minutes with batteries until generators start.
- **Backup generation:** Diesel or gas generators carry the full critical load for extended outages and are tested under load on a schedule.
- **Distribution:** Busway and rack power distribution units deliver clean power to each rack, metered so load is visible per row and per rack.
- **IT load:** GPU racks are far denser than traditional server racks, which is why AI halls are designed around rack power first.

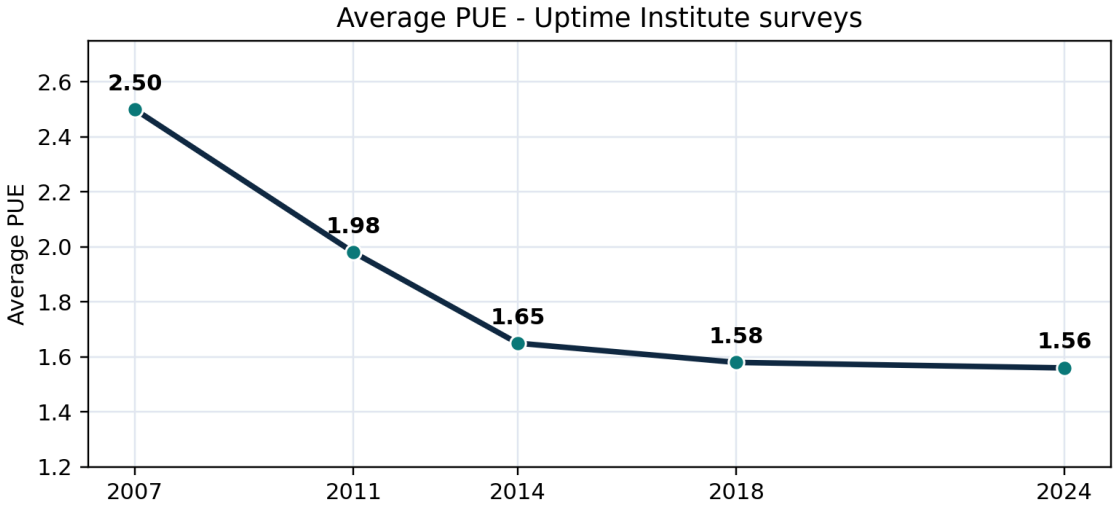
The heat chain

Nearly all electricity used by IT equipment becomes heat. Air cooling suits lower densities; direct liquid cooling and rear-door heat exchangers handle dense GPU racks by moving heat into water loops close to the chips, then to cooling distribution units, chillers or dry coolers, and finally out of the building.

Image is an illustration of a power room, not a specific site. Resilience terms such as N, N+1 and 2N describe how many independent units are installed beyond the number needed.

Efficiency: the average PUE has stalled at 1.56

Power Usage Effectiveness (PUE) is total facility power divided by IT equipment power. A PUE of 1.0 would mean every watt reaches the computers; 1.56 means that for every 100 watts of computing, another 56 watts go to cooling, power conversion and other overhead.





Why the average stopped improving

- Early gains came from cheap fixes: hot- and cold-aisle containment, blanking panels and variable-speed fans and drives.
- Much of the operating fleet is older and was built for air cooling at lower densities, so it cannot match new-build performance without major refits.
- New purpose-built facilities commonly design around 1.3, and leading hyperscale sites report figures near 1.1 - but there are not yet enough of them to move the global average.
- Climate matters: a facility without access to free cooling in a hot region cannot reach the same PUE as one in a cool region with the same equipment.

Beyond PUE

PUE says nothing about water, carbon or how hard the IT equipment is working. Read it alongside Water Usage Effectiveness (WUE), the carbon intensity of the local grid, and IT utilization. A half-empty hall can post a poor PUE no matter how good its plant is, because overhead is spread over too little computing.

Source: Uptime Institute Global Data Center Survey 2024 - average PUE 1.56, flat for a fifth consecutive year. Historical points: 2.5 (first average), 1.98 in 2011, 1.65 in 2014, 1.58 in 2018.

Facility profiles: Edge, Regional and Campus

The three profiles used in the dashboard. Choose by workload and latency need, not by size alone.

Profile	Best for	Typical character	Watch out for
Edge	Inference close to users, factories, hospitals and retail sites	Small footprint, fast deployment, low latency, limited on-site staff	Per-site cost, remote operations, physical security at many locations
Regional	Mixed cloud and AI inference for a metro or region	Mid-scale, balanced cost and resilience, good carrier choice	Power headroom for GPU growth; cooling refits as density rises
Campus	Large AI training and hyperscale cloud	Multiple halls on one site, dedicated substation, lowest unit cost at scale	Grid queue time, water and community impact, single-region risk

How the choice changes the design

Power and cooling

Edge sites favor self-contained units and air or small liquid loops. Regional sites mix air and liquid as densities climb. Campuses justify dedicated substations, large liquid-cooling plants and heat-rejection systems sized for the hottest week, not the average day.

Operations

Edge runs lights-out with remote hands. Regional sites keep a small resident team. Campuses run full shift teams with separate electrical, mechanical and IT operations, and formal change and incident processes like those on page 9.

A simple decision rule

- If users or machines cannot tolerate delay, start at the Edge and centralize only what latency allows.
- If the workload is training large models, it belongs on a Campus with secured power at scale.
- If neither is decisive, a Regional profile usually gives the best balance of cost, resilience and room to grow.

> **Planning note:** These profiles are planning categories, not industry standards. Any capacity, cost or PUE figure for a specific project must come from that project's own design and measurements.

Operations documents: the checklists that keep a hall safe

Templates to complete for a real facility. Blank boxes are intentional - a completed, dated copy is the record.

Daily rounds

[ ] Utility feeds and switchgear status normal

[ ] UPS load, battery state and alarms reviewed

[ ] Generator fuel level and start readiness

[ ] Cooling plant: chillers, pumps, temperatures

[ ] Hall temperatures and hot spots checked

[ ] Leaks, alarms and open doors investigated

[ ] Overnight events logged and handed over

Change control

[ ] Change described, with rollback plan

[ ] Risk and affected systems assessed

[ ] Approved by operations lead

[ ] Work window and notifications agreed

[ ] Result verified and record closed

Incident record

- **Time detected / source:** alarm, monitoring or report
- **Impact:** systems, customers, duration
- **Immediate action:** what stopped the harm
- **Root cause:** evidence-based, not assumed
- **Fix and prevention:** owner and due date

Energy log (weekly)

[ ] Total facility energy (kWh) recorded

[ ] IT energy (kWh) recorded

[ ] PUE calculated for the same period

[ ] Peak demand (MW) and when it occurred

[ ] Water use recorded where cooling uses water

[ ] Unusual load explained in one line

Maintenance rhythm

Cadence	Typical work
Weekly	Generator check, alarm review, filter and leak inspection
Monthly	Generator load test, UPS and battery inspection, thermal scan of halls
Quarterly	Cooling plant service, switchgear inspection, failover test planning
Annually	Full load-bank or integrated systems test, capacity and PUE review, document audit

Adapt intervals to the equipment makers' instructions and local regulations. Safety rules, permits and licensed-work requirements always override a generic checklist.

Sources, glossary and live-feed readiness

Sources - every figure in this pack traces to one of these

- **IEA - Energy and AI (April 2025).** Global use 415 TWh in 2024 (about 1.5%), base case 945 TWh by 2030, about 15% annual growth, accelerated servers about 30% per year, regional shares U.S. 45%, China 25%, Europe 15%. https://www.iea.org/reports/energy-and-ai/executive-summary
- **DOE / LBNL - 2024 Report on U.S. Data Center Energy Use.** U.S. use 176 TWh in 2023 (4.4%); projection 325-580 TWh by 2028 (6.7% to 12%).
- **DOE / LBNL - United States Data Center Energy Usage Report: 2025 Update.** U.S. use 192 TWh in 2024 (4.7%); 2030 reference case 649 TWh (11.8%), scenario range 521-843 TWh (9.5% to 15.3%). https://www.energy.gov/documents/united-states-data-center-energy-usage-report-2025-update
- **Uptime Institute - Global Data Center Survey 2024.** Average PUE 1.56 in 2024, flat for a fifth consecutive year; historical averages 1.98 (2011), 1.65 (2014), 1.58 (2018).

Glossary

Term	Meaning
PUE / WUE	Power / Water Usage Effectiveness - total facility use divided by IT use; closer to 1.0 is more efficient
TWh / MW / GW	Terawatt-hour (total energy over time); megawatt and gigawatt (power at a moment)
UPS / DCIM	Uninterruptible power supply; Data Center Infrastructure Management software
GPU / Inference	Graphics processing unit used for AI; inference is running a trained model for users
N+1 / 2N	Redundancy: one spare unit beyond need / a fully duplicated system

Live-feed readiness - before any live view can be claimed

No live facility feed is connected to this pack or to the dashboard. A real live view needs all of the following from the facility owner - nothing here can substitute for them:

[ ] Feed owner and system name (for example a DCIM or building management system)

[ ] Endpoint address reachable without exposing secrets on a public page, or a proxy the owner controls

[ ] Metric list and format: power, temperature, rack and alarm data, with units and update interval

[ ] Written permission to display the data publicly

When those are provided, the dashboard can be wired to the real feed and this page updated with the connection details and date.

Company finance and business employment links

Official starting points for funding an AI data center company and hiring its team. Every link was checked live on 17 October 2026; programs and terms change, so confirm details on the site itself.

Company finance

Resource	What it is for
SBA Loans - U.S. Small Business Administration https://www.sba.gov/loans/	The main U.S. small-business loans: 7(a) for general financing, 504 for long-term fixed-rate financing of land, buildings and equipment, and microloans of $50,000 or less. Start here for eligibility.
DOE Loan Programs Office - Office of Energy Dominance Financing https://www.energy.gov/lpo/loan-programs-office	Federal financing for large energy projects that support grid reliability and energy security - the scale relevant to a data center's power side, such as generation and grid upgrades.
Grants.gov https://www.grants.gov/	The single search point for federal grants across U.S. agencies, for organizations. Federal agencies do not publish personal financial assistance here.
IRS - Small Business and Self-Employed Tax Center https://www.irs.gov/businesses/small-businesses-self-employed	Business taxes in one place: Employer ID Numbers (EINs), forms, employment taxes, tax credits, deducting expenses and recordkeeping.

Business employment and hiring

Resource	What it is for
SBA - Hire and Manage Employees https://www.sba.gov/business-guide/manage-your-business/hire-manage-employees	Step-by-step employer setup: get an EIN, set up payroll and withholding, complete W-4 and I-9 forms, and cover required benefits - Social Security taxes, workers compensation, unemployment insurance.
U.S. Department of Labor - Hiring https://www.dol.gov/general/topic/hiring?lang=en	Federal hiring guidance: recruiting help through American Job Centers and the CareerOneStop Business Center, Form I-9 requirements, and hiring veterans and foreign workers.
BLS - Occupational Outlook Handbook: Computer and Information Technology https://www.bls.gov/ooh/computer-and-information-technology/home.htm/	Bureau of Labor Statistics pay, job counts and outlook for the IT occupations a data center hires - benchmark wages and check which roles are growing before setting salaries.
CareerOneStop https://www.careeronestop.org/	The Department of Labor career portal: job search, occupation profiles, training finder and local help through American Job Centers - for job seekers and employers alike.

> **How to use this page:** Check SBA loan eligibility first, search Grants.gov for a matching grant, and talk to the DOE loan office early for a large power project - grid lead times are the longest part of any build. For hiring, set pay from the BLS figures and set up payroll with the SBA steps.

Public information links, not financial or legal advice. This pack holds no live company financial data and no job listings of its own.